

Economics of Social Media Fake Accounts

Zihong Huang,^a De Liu^{b,*}

^aRawls College of Business, Texas Tech University, Lubbock, Texas 79409; ^bCarlson School of Management, University of Minnesota, Minneapolis, Minnesota 55455

*Corresponding author

Contact: zihong.huang@ttu.edu,  <https://orcid.org/0000-0003-3667-9329> (ZH); deliu@umn.edu,  <https://orcid.org/0000-0002-9430-9732> (DL)

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Abstract. Amid the rise of the influencer economy, fake social media accounts have become prevalent on many social media platforms. Yet the problem of fake accounts is still poorly understood, and so is the effectiveness of coping strategies. This research models the ecosystem of fake accounts in an influencer economy and obtains insights on fake account purchasing behaviors, the impact of antifake efforts, and the roles of various contextual factors. We show that as the antifake effort increases, the equilibrium may transition from a “pooling” equilibrium, where a low-quality influencer buys fake accounts to mimic a high-quality one, to a “costly separating” equilibrium, where a high-quality influencer may buy fake accounts to prevent mimicry from a low-quality influencer, and to a “naturally separating” equilibrium where low- and high-quality influencers are separated without buying fake accounts. We find that increasing antifake efforts and increasing social media literacy may sometimes result in more fake accounts. A purely profit-driven platform always prefers a pooling equilibrium with zero antifake effort. As a platform puts more weight on consumer welfare, it may exert a positive effort to induce a separating equilibrium, but the platform’s preferred antifake effort tends to be lower than that of consumers. We also find that the platform sometimes prefers a lower social media literacy and a lower fake account base price, whereas consumers prefer the opposite. In contrast, improving the antifake technology level can benefit both the platform and consumers. Our main insights are applicable to scenarios with more influencer types and repeated interactions.

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1. Introduction

On October 16, 2019, a popular microblogger with 3.8 million followers on Weibo, one of the largest microblogging platforms in China, posted an advertisement. Within 50 minutes, the advertisement garnered 121,000 views, thousands of likes, and hundreds of comments and shares. The advertiser was thrilled to see the response but surprised by the number of conversions: zero! It turned out that the microblog was infested with fake followers. This incidence is not an aberration: Facebook, Instagram, and Twitter have all been reported struggling with fake account problems (Confessore et al. 2018, Moore and Murphy 2019, Dang and Paul 2022).

By “fake accounts,” we mean social media accounts designed to impersonate real users with fake personal information and/or behaviors. A majority of fake accounts are automated (or “bots”), whereas fake accounts may also be created and operated by real humans (Nicas 2020). Fake accounts are created for several reasons. The most common use of fake accounts is

to boost influence: social media influencers may buy fake accounts to make their accounts or social media posts appear more influential, which can result in more followers and higher advertising income. Some users may create fake accounts to obtain perks such as signup bonuses and coupons. Others may use fake accounts to spread phishing, scams, and malware. Yet others may use fake accounts to sway public opinion or election outcomes. This paper focuses on influence-boosting fake accounts because of their prevalence.

The demand for influence-boosting fake accounts has surged in the influencer economy (Confessore et al. 2018, Federal Trade Commission 2019), in which large and small social media influencers can get paid for each piece of promotional content (e.g., sponsored posts and product endorsements) they share, depending on their reach (e.g., number of followers) and engagement (e.g., clicks and likes) metrics.¹ As such, influencers have strong incentives to use fake accounts to artificially boost their reach and engagement metrics. The link

between the influencer economy and fake accounts is highlighted in the widely publicized case of the *Federal Trade Commission (FTC) vs. Devumi* in 2019. Devumi is a company that made millions of dollars by manufacturing and selling fake accounts and associated services to actors, athletes, musicians, and other high-profile individuals who wanted to appear more popular and influential online (Confessore et al. 2018). Although the FTC imposed a fine of \$2.5 million on Devumi with the intent of deterring future fake account trading, the fake account problem has never abated. For example, in the first quarter of 2022, Facebook shut down 1.6 billion fake accounts and estimated that no less than 5% of the Facebook users were still fake after the removal (Warwick 2022).

Because fake accounts can result in wasteful marketing expenditures and mislead social media users, social media platforms have already begun to tackle the fake account problem. A major tool used for fighting fake accounts is automated fake account detection and prevention. For example, Facebook uses machine learning to detect and block fake accounts both before and after they come alive (Hao 2020). Social media platforms also use user verification technologies such as reCAPTCHA and two-factor authentication to deter automated fake account creation. Such platform-led *antifake efforts* face considerable technical challenges, however. Not only is there a wide range of user behaviors, making it nearly impossible to automatically differentiate between real and fake accounts (Nicas 2020), but also fake account providers are increasingly adept at emulating genuine users with advanced AI techniques (a recent report found computer-generated images in fake LinkedIn profiles; see Robins 2022). Given these technical challenges, it is unclear how platform-led antifake efforts may affect the prevalence of fake accounts.

There is also a concern about whether social media platforms can be trusted to resolve the issue of fake accounts on their own. Social media platforms, because of their inherent interest in maintaining a large number of active users, may be reluctant to reveal the extent of fake accounts or remove them. Moreover, antifake efforts such as fake account detection and user verification tend to inconvenience legitimate users. Heightened user verification aimed at preventing fake accounts can frustrate legitimate users, potentially driving them away from the platform (Anderson 2024). In general, we do not yet know whether the platform can be expected to exert the level of antifake efforts that is ideal for consumers.

In addition to platform-led antifake efforts, a few contextual factors may potentially impact the prevalence of fake accounts. For example, schools and online education institutions have taken initiatives to improve *social media literacy* among both young and adult social media users (Al Zou'bi 2022). With enhanced social media

literacy, users can better discern the veracity of information on social media, which may reduce the influence of fake accounts on consumer decisions (Polanco-Levicán and Salvo-Garrido 2022). Meanwhile, regulators and government agencies can also use tools such as regulations and fines to make it more costly to operate and trade fake accounts. To develop effective strategies for tackling the fake account problem, we need to understand how such factors impact the prevalence of fake accounts and consumer welfare and the profitability of social media platforms.

Our current understanding of the fake account phenomenon remains limited because of the scarcity of reliable data on fake accounts. For example, we do not know what types of influencers are more likely to buy fake accounts or if it is always better to have fewer fake accounts. Existing studies of fake accounts primarily focus on examining the activities of fake accounts and developing detection techniques (Raturi 2018, Yuan et al. 2019). Numerous questions surrounding the fake account phenomenon, including those highlighted above, remain unanswered. This suggests a need for a systematic examination of the fake account phenomenon so that multiple related questions can be answered holistically.

To address this need, we build a stylized game-theoretical model of fake accounts in the context of an influencer economy. This model comprises several stakeholders, including consumers, a representative influencer, an advertiser, and a social media platform. The influencer can be either high quality or low quality. A proportion of consumers are “informed” about the influencer’s quality, whereas the rest are “uninformed.” Informed consumers make their following decisions first. Uninformed consumers make their following decisions after seeing the influencer’s follower count, which could be inflated by fake accounts. In this economy, the advertiser reaches consumers through the influencer, and both the influencer and the platform get a share of the advertising revenue. The platform can mount an antifake effort that increases the price of fake accounts, but such an effort inevitably also increases the nuisance cost of consumers. The *antifake technology level* governs the extent to which the antifake effort can raise the price of fake accounts without imposing much nuisance cost on consumers. In our model, the platform maximizes a weighted sum of profits and consumer welfare. When the platform assigns zero weight to consumer welfare, it is *purely profit driven*. As the weight increases, the platform becomes more *consumer oriented*. We use this model to address a host of questions, such as:

1. What is the social media influencer’s equilibrium fake account purchasing behavior?
2. How does the platform’s antifake effort affect the number of fake accounts, the platform’s profits, and consumer welfare? Is the platform’s preferred antifake effort aligned with consumers?

3. How do contextual factors such as the antifake technology level, social media literacy (as measured by the proportion of informed consumers), and the base price of fake accounts affect the equilibrium, the platform's profits, and consumer welfare?

Our analyses show that there is a *pooling* equilibrium where a low-quality influencer (*L*-type for short) *offensively* purchases fake accounts to mimic a high-quality influencer (*H*-type), a *costly separating* equilibrium, where an *H*-type *defensively* purchases fake accounts to prevent an *L*-type from mimicking, and a *naturally separating* equilibrium where the two types of influencers separate without needing to purchase fake accounts. As the platform's antifake effort increases, the equilibrium generally transitions from a pooling, to a costly separating, and then to a naturally separating equilibrium. Interestingly, the number of fake accounts may sometimes increase with the antifake effort, especially in the pooling equilibrium. This is because under a pooling equilibrium, a higher antifake effort results in a larger gap in the follower counts of both the *H*- and *L*-type, forcing the *L*-type to purchase more fake accounts despite higher costs. We also observe that the number of fake accounts can jump significantly as the system transitions from a pooling to a separating equilibrium. This is because an *L*-type is willing to pay more to be seen as an *H*-type in a separating equilibrium than as an average type in a pooling equilibrium.

We show that a purely profit-driven platform always prefers a zero antifake effort, which results in a pooling equilibrium. The reason is that a pooling equilibrium can result in more uninformed followers and higher advertising revenue. As the platform becomes more consumer oriented, it may exert a positive effort to induce a costly separating or a naturally separating equilibrium. In general, though, the platform's optimal antifake effort tends to be lower than what is optimal for consumers.

Moreover, whereas consumers benefit from a higher level of social media literacy and a higher fake account base price, a sufficiently profit-focused platform may prefer the opposite. We also find that under a pooling equilibrium, the number of fake accounts is unaffected by the base price of fake accounts because under such an equilibrium, the *L*-type must make up for the follower gap regardless of the price of fake accounts. This suggests that the social media platform may lack interest in improving social media literacy or raising the price of fake accounts, although such measures can improve consumer welfare. In contrast, both the platform and consumers may benefit from improving antifake technology (under a separating equilibrium).

In addition, we show that key insights from the one-shot game with two types of influencers could carry over a repeated setting and a model with more influencer types.

2. Related Literature

To our knowledge, the fake social media account problem has not been formally modeled in the literature. However, the literature has studied several other types of deceptive/manipulative behaviors in commerce and advertising contexts, including deceptive advertising (Piccolo et al. 2018), fake reviews, fake sales (Jin et al. 2023), and click fraud (Wilbur and Zhu 2009). In general, our problem and focus are quite different, but there are similarities in the analytical framework. Below, we discuss the relationship between our research and prior studies of deceptive/manipulative behaviors from three aspects: equilibrium behaviors, coping strategies, and welfare implications.

First, our paper is connected to several studies of equilibrium deceptive/manipulative behaviors that also use the signaling model as the analytical framework. In general, the stream on deceptive advertising as well as fake sales usually studies a game in which sellers compete for buyers using deceptive tactics such as false advertising, fake purchases, fake reviews, and so on, along with pricing decisions. In contrast, influencers in our setting have no pricing decisions; they only need to decide how many fake accounts to purchase to influence consumer and advertiser perceptions of them. Furthermore, the previous studies mainly focus on one type of equilibrium. For instance, Piccolo et al. (2018) characterize a class of pooling equilibria where the *L*-type sellers deceive a buyer. Similarly, Mayzlin (2006) also finds a pooling equilibrium in which sellers with inferior products lie. In contrast, another paper in the same stream focuses on a separating equilibrium (Corts 2013). Recently, Chen and Papanastasiou (2021) studied a game in which the seller manipulates the buyers' beliefs with fake purchases. They found that a bad (i.e., *L*-type) seller may or may not engage in manipulation (i.e., make fake purchases), whereas a good (*H*-type) one never manipulates. We, on the other hand, find a different pattern; not only does the *L*-type influencer conduct fake account purchasing in the pooling equilibrium, but the *H*-type also manipulates (i.e., buys fake accounts) in the costly separating equilibrium as well. In addition, we also identify a naturally separating equilibrium in which neither type buys fake accounts.

Second, our paper is also related to a small literature set on the effectiveness of antifake strategies. This literature has studied the strategies for helping consumers learn the true quality of products through information disclosure (Che and Hörner 2018, Papanastasiou et al. 2018, Pennycook et al. 2020) and penalizing the information producers for their manipulative behaviors (Corts 2014, Papanastasiou 2020). In particular, Chen and Papanastasiou (2021) study the detection-and-removal strategy against seller manipulation (e.g., via fake purchases and reviews) and observe that more intensive

detection and removal may lead to more seller manipulation because it increases consumers' trust, which further leads to higher equilibrium prices and greater seller manipulation. We also find that antifake efforts may sometimes lead to more fake accounts, but for a different reason: it can increase the gap between H - and L -type influencers, which forces the L -type to buy more fake accounts to make up for the gap. Our model of the antifake efforts is also different: they increase the cost of fake accounts but also increase the nuisance costs of consumers. Importantly, we gain insights into other interventions such as increasing the level of antifake technology and social media literacy, which are new to this literature.

Third, our study is related to research on the welfare effects of deceptive behaviors. Piccolo et al. (2018) examine how consumer welfare is affected by sellers' deceptive strategies. They suggested that consumer welfare could be higher under the equilibrium with sellers' deceptive advertising. Jin et al. (2023) study the impact of brushing (e.g., fake sales) on consumer welfare and find that brushing can either improve or hurt consumer welfare. Our work on consumer welfare is closest to that of Chen and Papanastasiou (2021), which suggests that seller and consumer welfare can be maximized at an intermediate level of antifake effort by the platform (e.g., detecting fake reviews) or the government (e.g., law enforcement against fake product endorsements). Different from Chen and Papanastasiou's work, our study is set in the context of an influencer economy rather than an e-commerce setting. We study not only the platform's antifake effort from the consumer welfare point of view but also the welfare impact of other interventions, such as increasing social media literacy, increasing fake account base price (which has a similar interpretation as the government's antifake effort), and increasing effectiveness of antifake technology.

Finally, our paper should be contrasted with the study of click fraud by advertisers in the context of search engine keyword auctions by Wilbur and Zhu (2009). Their focus is on the unfair competition between advertisers in an auction context and its impact on search engine revenue. Their game has a very different structure from ours. In addition, they do not study search engine strategies for coping with click fraud or consumer welfare implications.

3. The Model

We model fake accounts in the context of influencer marketing, where advertisers pay influencers on social media platforms to promote their brands/products. The ecosystem consists of a social media platform, a representative influencer, a unit mass of consumers, and a representative advertiser. Figure 1 depicts their relationships, which we discuss further below. The notations used in this paper are summarized in Table 1.

3.1. Platform

The social media platform facilitates influencer marketing campaigns, including providing tools for advertisers and influencers to find each other, launch and monitor influencer marketing campaigns, and make payments for such campaigns.² In return for such services, the platform receives a share of the revenue generated by influencer marketing campaigns.

The platform chooses the level of antifake efforts d ($d \geq 0$) (effort for short), which may include efforts spent on detecting, removing, and preventing fake accounts. For example, the platform may use machine learning to detect fake accounts based on user attributes and behaviors. Such detection can be used to prevent new fake accounts and remove existing ones. It may also deploy technologies such as reCAPTCHA to make it harder to register and operate fake accounts in automated ways. The platform incurs a cost of $(\gamma/2)d^2$ ($\gamma > 0$) for its antifake effort. The quadratic functional form reflects diminishing returns of such antifake efforts.

The platform's antifake efforts have two effects. First, they increase the unit price of fake accounts p_f , which is given by

$$p_f = p_0 + \frac{1}{1 - \tau} d, \quad (1)$$

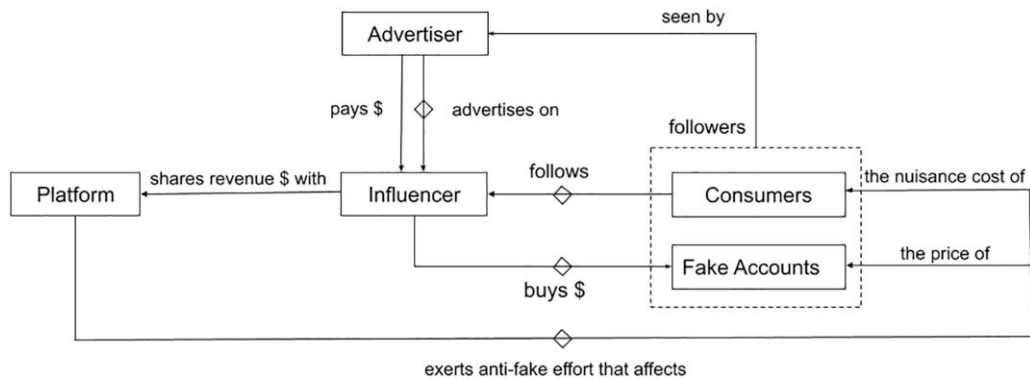
where p_0 is the base price of fake accounts, and τ ($1 > \tau \geq 0$) is the antifake technology level. The base price of fake accounts reflects the expenses associated with operating a basic fake account. These costs may be influenced by factors such as the time, effort, and technologies used for creating and maintaining such accounts, as well as the legal risk faced by fake account operators. The antifake technology level reflects the effectiveness of current antifake technology, which may be understood as the accuracy of machine learning algorithms for detecting fake accounts and the discriminative power of antifake tools such as reCAPTCHA. This formulation captures the notion that both antifake efforts and more effective antifake technology can increase the costs of operating fake accounts.

Second, the platform's antifake efforts increase the nuisance costs for legitimate users. For example, heightened antifake efforts may result in legitimate users answering reCAPTCHA questions more frequently, resulting in an increasing likelihood of being misclassified as fake accounts. In general, we assume a consumer incurs a cost c from following the influencer

$$c = c_0 + c_1(1 - \tau)d, \quad (2)$$

where c_0 is the opportunity cost of time, and $c_1(1 - \tau)d$ is the nuisance cost imposed by the platform's antifake efforts. By this formulation, as the level of antifake technology τ increases, consumers' nuisance cost decreases (e.g., they are less likely mistaken as fake accounts).

Figure 1. Model Sketch



We assume the platform maximizes a weighted sum of its profit π_p and consumer welfare U ,

$$\Pi_p = (1 - w)\pi_p + wU, \quad w \in [0, 1], \quad (3)$$

where the parameter w measures the platform's *consumer orientation*. If $w = 0$, the platform is *purely profit driven*. As w increases, the platform becomes more *consumer oriented*. The literature shows that firms have

goals beyond profit seeking, such as social responsibility goals (Lankoski and Smith 2018).

3.2. Influencer

The influencer can be realized as one of the two types, H -type or L -type, with content qualities q_H and q_L ($q_H > q_L$), respectively. The probability of the influencer being the H -type is ρ . For simplicity, we assume the influencer produces one unit of content. Following prior

Table 1. Notation

Panel A: Decision variables	
Variable	Description
d	The platform's antifake effort
x	The number of fake accounts purchased by the influencer
n_{in}, n_{un}	The number of informed and uninformed consumers that follow the influencer, respectively
n_2, n	The number of early and total followers (including both real and fake accounts) of the influencer, respectively
$n_{r,ia}, E[n_{r,ua}], E[n_r]$	The expected number of real followers facing an informed, uninformed, and average advertiser, respectively
Panel B: Parameters	
Parameter	Description
c_o	A consumer's opportunity cost
c_1	The nuisance cost coefficient
l	The proportion of informed consumers and the probability of the advertiser being informed (also the level of social media literacy)
p_f, p_0	The unit price and base price of fake accounts, respectively
q_H, q_L	The H -type and L -type influencer's content quality, respectively
γ	The cost coefficient for the platform's antifake effort
$\theta \in [0, 1]$	Consumers' valuation for the influencer's content quality
$\lambda_i, \lambda_p, \lambda_a$	The bargaining power of the influencer, the platform, and the advertiser, respectively
μ	The revenue generated by advertising to a real consumer
ρ	The probability of drawing an H -type influencer
$\tau \in [0, 1]$	The platform's antifake technology level
$w \in [0, 1]$	The platform's consumer orientation
Panel C: Outcome variables	
Variable	Description
u_i	Consumer i 's expected utility
U	Consumer welfare
π_a, π_i, π_p	The expected profit of the advertiser, the influencer, and the platform, respectively
Π_p	The platform's expected utility: a weighted sum of its profit and consumer welfare

literature (Shin 2017), we normalize the cost of producing the content to zero. The influencer chooses the number of fake accounts x to maximize its profit π_i (which we will define shortly).

3.3. Consumers

There is a unit mass of risk-neutral consumers. Each consumer decides whether to follow the influencer.³ A consumer's utility from following an influencer is given by

$$u(\theta, q) = \theta q - c, \quad (4)$$

where q is the quality of the content, θ is the consumer's valuation for quality, and c is the cost as defined in (2). We assume θ is random and uniformly distributed on $[0, 1]$.

We further assume that some consumers are better informed than others. For simplicity, we assume a proportion l of consumers are *informed*; that is, they know the influencer's true quality.⁴ Such consumers tend to have extensive experience with social media and a stronger ability to tell an influencer's quality. The remaining consumers are *uninformed*: they only know the distribution of the influencer's quality but can observe the influencer's existing followers and form an updated belief about the influencer's quality.⁵ Our assumption of uninformed consumers using the influencer's followers to infer quality is supported by prior empirical findings. For example, research shows consumers perceived the influencers with a higher number of followers as being more attractive (Jin and Phua 2014), trustworthy (Jin and Phua 2014), and likable (De Veirman et al. 2017).

We interpret the proportion of informed consumers l as the level of *social media literacy*: the higher the social media literacy, the more consumers are informed and unaffected by fake account-based manipulations.

3.4. Advertiser

There is one representative advertiser.⁶ The advertiser promotes their brand by asking the influencer to share a sponsored post or a product endorsement among the influencer's followers. The advertiser derives a unit revenue μ from advertising to a *real* consumer. The advertiser could be one of two types: *informed* or *uninformed*. For simplicity, we assume the probability of the advertiser being informed is also l , the same as the proportion of informed consumers. We conduct an analysis with a more general specification in which the advertiser can be informed with a higher probability, and our findings are consistent.

The influencer's type, consumer valuation θ , and advertiser's type are all private information.

3.5. Revenue Sharing

The total advertising revenue generated by the system is $\mu E[n_r]$, where μ is the revenue generated by advertising to a real consumer, and $E[n_r]$ is the expected number

of real consumers reached. Consistent with common practice in the influencer marketing industry, we assume this revenue is shared between the three parties: the influencer, the advertiser, and the platform. Denote $\lambda_i \in (0, 1)$ and $\lambda_p \in (0, 1)$ as the influencer's and the platform's shares of revenue, respectively. We denote $\lambda_a \equiv 1 - \lambda_i - \lambda_p$, $\lambda_a \in (0, 1)$ as the advertiser's share of the revenue. The parameters λ_i and λ_p are exogenously given and reflect the two parties' relative *bargaining power*.

Given the revenue sharing scheme, the three parties' profits are, respectively,

$$\pi_i = \lambda_i \mu E[n_r] - p_f x \quad (5)$$

$$\pi_p = \lambda_p \mu E[n_r] - \frac{\gamma}{2} d^2 \quad (6)$$

$$\pi_a = \lambda_a \mu E[n_r]. \quad (7)$$

We note that because the advertiser has no cost, it will always participate in the partnership as long as the platform and the influencer are willing to.

We make the following two assumptions about model parameters. Assumption 1 posits that consumers with the highest valuation for quality (i.e., $\theta = 1$) derive positive utility from following an L -type influencer. Otherwise, the L -type cannot attract any informed followers, and the problem may degenerate. Assumption 2 ensures that the influencer has a high enough bargaining power so that there is an incentive to participate in the revenue sharing.

Assumption 1. $q_L > c$.

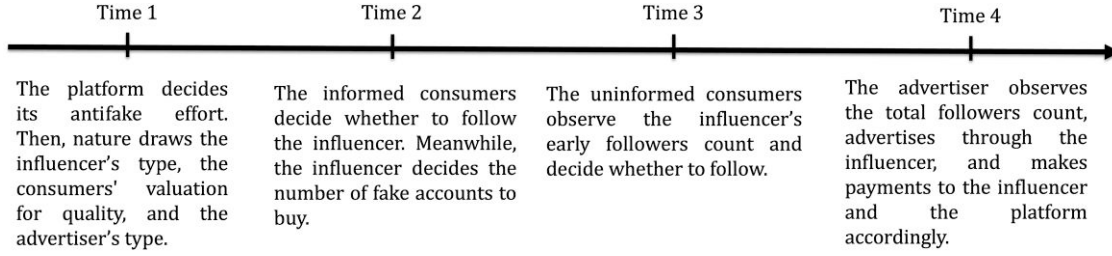
Assumption 2. $\lambda_i > \frac{p_f}{\mu + p_f}$.

3.6. Game Timeline

The timeline of the game (Figure 2) is as follows. At time 1, the platform decides its antifake effort d . Then, nature draws the influencer's type, the consumers' valuation for quality θ , and the advertiser's type. At time 2, informed consumers decide whether to follow the influencer, and the influencer decides the number of fake accounts x to buy. Now, the influencer has n_2 *early followers*, which include n_{in} informed consumers and x fake accounts. At time 3, uninformed consumers observe the influencer's early followers n_2 and decide whether to follow. After the uninformed consumers' decisions, the influencer has n total followers, which include n_2 early followers and n_{un} uninformed consumers. At time 4, the advertiser observes the total follower count n , advertises through the influencer, and makes payments to the influencer and the platform accordingly.

This game timeline captures a few key aspects of different stakeholders' decision environments and the main effects of their decisions. First, we assume informed consumers make their following decisions

Figure 2. Game Timeline



before uninformed ones. This is because uninformed consumers rely on the influencer's follower count to infer quality, and thus, it is natural for them to wait for informed consumers' decisions. Second, we assume the influencer purchases fake accounts before uninformed consumers make their decisions. This is to ensure the influencer has a chance to influence their following decisions. Third, we assume the influencer's fake account purchase and the informed consumers' following decisions occur simultaneously because these decisions are independent of each other, and the model would remain the same if they occur sequentially. Fourth, we assume the platform's antifake effort occurs before the influencer's fake account purchase. This ensures the former can influence the price of fake accounts. It also captures the notion that for a fake account to work, it must survive the platform's antifake effort. Finally, we assume that the advertiser moves last because advertisers often begin to advertise with an influencer when they are popular enough, at which point the influencer has already accumulated followers.

4. Equilibrium Analysis

4.1. Preliminaries

Given the Consumer Utility (4), the number of informed followers for the t -type influencer is

$$n_{in}^t = l \left(1 - \frac{c}{q_t} \right), \quad t \in \{H, L\}. \quad (8)$$

It is easy to see that the H -type has more informed followers than the L -type (i.e., $n_{in}^H > n_{in}^L$).

At time 3, the number of uninformed consumers following the t -type influencer is

$$n_{un}^t = (1 - l) \left(1 - \frac{c}{E[q | n_2^t]} \right), \quad t \in \{H, L\}, \quad (9)$$

where n_2^t is the number of early followers for the t -type, $E[q | n_2^t] = \Pr(H | n_2^t)q_H + \Pr(L | n_2^t)q_L$ is the expected quality of the influencer conditional on the number of early followers n_2^t , and $\Pr(t | \cdot), t \in \{H, L\}$ is the conditional probability of the influencer being the t -type.

At time 4, an informed advertiser knows the influencer's type and can correctly anticipate the number of

real followers, which is

$$n_{r,ia}^t = n_{in}^t + n_{un}^t, \quad t \in \{H, L\}. \quad (10)$$

An uninformed advertiser forms an expectation of the number of real followers:

$$E[n_{r,ua}] = \Pr(H | n)(n_{in}^H + n_{un}^H) + \Pr(L | n)(n_{in}^L + n_{un}^L), \quad (11)$$

where $\Pr(t | n)$ is the probability of the influencer being t -type, conditional on the number of total followers n . Therefore, from the perspective of an t -type influencer, the expected number of real followers that the advertiser would pay for is

$$E[n_r^t] = l n_{r,ia}^t + (1 - l)E[n_{r,ua}]. \quad (12)$$

4.2. Influencer's Equilibrium Decision

The game between the influencer, consumers, and the advertiser is a variation of the signaling game where the influencer attempts to signal their type to both uninformed consumers and advertisers. Because the number of total followers carries the same information as the number of early followers,⁷ without loss of generality, we use the number of early followers as a signal for both uninformed consumers and advertisers.

A strategy profile of this game can be stated as (n_2^H, n_2^L) , that is, the number of early followers for the H - and L -type influencers, respectively. It can be equivalently stated as (x_H, x_L) , that is, the number of fake accounts purchased, given that $n_2^t = n_{in}^t + x_t$ ($t \in \{H, L\}$). Following the signaling game literature, we classify the equilibria as pooling and separating equilibria. If the number of early followers is the same for the two types of influencers, it is a *pooling equilibrium*; otherwise, it is a *separating equilibrium*. We further classify the separating equilibrium into two kinds: (a) a *naturally separating equilibrium* where neither type of influencer purchases fake accounts, and (b) a *costly separating equilibrium* where at least one type purchases fake accounts. A similar distinction has been made by Guo et al. (2017) in the context of corporate social responsibility.

The signaling game tends to have multiple perfect Bayesian equilibria (PBEs). A common way to refine the

equilibrium is the *undefeated equilibrium refinement* proposed by Mailath et al. (1993). One benefit of the undefeated refinement is to avoid the global consistency issue associated with the *intuitive criterion* (Mailath et al. 1993), another popular refinement strategy. Moreover, undefeated refinement also tends to select a unique PBE in signaling games. Therefore, we adopt the undefeated refinement in this research.⁸

In the following subsections, we first apply the undefeated refinement within each equilibrium type to obtain a *locally undefeated* equilibrium and then apply it across equilibrium types to obtain the *globally undefeated* equilibrium.

4.2.1. Pooling Equilibrium. In the pooling equilibrium, H -type and L -type have the same number of early followers. We obtain a range of pooling PBEs with different levels of early followers. However, all higher-level pooling equilibria are defeated by the lowest-level one. The latter is the only locally undefeated pooling equilibrium as stated in the following lemma.

Lemma 1 (Pooling). *A strategy profile $(n_2^H, n_2^L) = (n_{in}^H, n_{in}^L)$ with the supporting beliefs*

$$P(H | n_2) = \begin{cases} 0, & \text{if } n_2 < n_{in}^H \\ \rho, & \text{if } n_2 = n_{in}^H \\ p \in [0, \bar{p}_1], & \text{if } n_{in}^H < n_2 < \bar{n}_2^{pool} \end{cases}$$

is the unique locally undefeated pooling equilibrium if and only if

$$d \leq (1 - \tau)(\eta_1 - p_0) \equiv d_1, \quad (13)$$

where $\bar{p}_1 \in [\rho, 1]$ and $\bar{n}_2^{pool}$ are given in Online Appendix A.1, and $\eta_1 \equiv \lambda_i \mu(1 - l)[(q_H(E[q] - q_L))/(E[q](q_H - q_L)) + \rho]$.

Lemma 1 describes an equilibrium where the L -type purchases fake accounts, whereas the H -type does not. In this equilibrium, the L -type purchases fake accounts to mimic the H -type in early followers so that they look identical to uninformed consumers and the uninformed advertiser. We call such purchasing *offensive purchasing*. Specifically, L -type's offensive purchasing x_L^{pool} is

$$x_L^{pool} = n_{in}^H - n_{in}^L. \quad (14)$$

The equilibrium belief holds that anyone who deviates to a lower follower count $n_2 < n_2^H$ (achievable only by an L -type) must be an L -type, whereas anyone who deviates to a higher follower count $n_2 > n_2^H$ is deemed as an H -type with probability $p \in [0, \bar{p}_1]$. The belief p is capped because an overly favorable belief would make deviations to higher follower counts profitable.

Condition (13) is derived from the L -type's incentive compatibility (IC) condition, ensuring that the L -type has no incentive to deviate. To understand this

condition, we rewrite (13) as

$$p_f = p_0 + \frac{1}{1 - \tau} d \leq \eta_1. \quad (15)$$

The left-hand side is the unit price of fake accounts, and the right-hand side can be interpreted as the revenue gain per fake account for the L -type to stay in the pooling equilibrium, compared with purchasing nothing and being treated as an L -type, their best alternative.

4.2.2. Separating Equilibrium. We describe the costly separating and naturally separating equilibrium schemes in the next two lemmas.

Lemma 2 (Costly Separating). *A strategy profile $(n_2^H, n_2^L) = (n_2^{csep}, n_{in}^L)$ with the support beliefs*

$$P(H | n_2) = \begin{cases} 0, & \text{if } n_2 < n_{in}^H \\ p \in [0, \bar{p}_2) & \text{if } n_{in}^H \leq n_2 < n_2^{csep} \\ 1 & \text{if } n_2 = n_2^{csep} \\ p \in [0, 1], & \text{if } n_2^{csep} < n_2 < \bar{n}_2^{csep} \end{cases}$$

is the unique locally undefeated costly separating equilibrium if and only if

$$d \leq (1 - \tau)(\eta_2 - p_0) \equiv d_2, \quad (16)$$

where $\bar{n}_2^{csep}$ is defined in Online Appendix A.2, $\eta_2 \equiv (1/l)\lambda_i \mu(1 - l^2)$, and

$$n_2^{csep} = \frac{1}{p_f} \lambda_i \mu [(1 - l)(n_{in}^H - n_{in}^L) + (n_{un}^H - n_{un}^L)] + n_{in}^L \quad (17)$$

$$d_1 < d_2. \quad (18)$$

Lemma 2 describes an equilibrium where the H -type purchases fake accounts to deter the L -type from mimicking. We call such purchasing *defensive purchasing*. Condition (16) is derived from the IC condition for the L -type, which ensures that the L -type will mimic the H -type if the latter purchases nothing. To further understand this condition, we rewrite it as

$$p_f = p_0 + \frac{1}{1 - \tau} d \leq \eta_2. \quad (19)$$

The left-hand side is the unit price of fake accounts, and the right-hand side η_2 can be interpreted as the revenue gain per fake account for the L -type to deviate from not buying to buying. By ensuring the L -type will deviate when the H -type buys nothing, we expect the H -type to purchase to maintain the separating equilibrium.

The H -type's equilibrium defensive purchase

$$x_H^{csep} = n_2^{csep} - n_{in}^H = \frac{1}{p_f} [\lambda_i \mu(1 - l^2) - p_f l] \left(\frac{c}{q_L} - \frac{c}{q_H} \right) \quad (20)$$

is the difference between their number of informed followers and the highest follower count the L -type is

willing to mimic. The term $(c/q_L) - (c/q_H)$ is the gap between the H -type and the L -type's real follower counts under the separating equilibrium. An L -type's gain from mimicking a nonpurchasing H -type is $\lambda_i \mu(1 - l^2)$ times this gap, noting that $\lambda_i \mu(1 - l^2)$ is the influencer's share of the advertising revenue per consumer. The L -type's cost of mimicking is $p_f l$ times the gap, noting that they only need to fill the gap in the informed followers, which is l proportion of all real followers. x_H^{sep} is chosen such that L -type is break even after having to purchase x_H^{sep} additional fake accounts.

The fact that H -type may purchase fake accounts in equilibrium is interesting in light of the literature's nearly exclusive focus on the L -type's deceptive behaviors (Piccolo et al. 2018, Chen and Papanastasiou 2021). In our study, the H -type may also buy fake accounts to deter the L -type's mimicry; even doing so will not fool the advertiser. Lemma 2 highlights that the H -type influencer may buy fake accounts not to deceive but to establish their superiority. This finding is consistent with the casual observation that high-status influencers frequently purchase fake accounts, as seen in the Devumi case (Mekuli 2021).

Lemma 3 (Naturally Separating). *A strategy profile $(n_2^H, n_2^L) = (n_{in}^H, n_{in}^L)$ with the support beliefs*

$$P(H | n_2) = \begin{cases} 0, & \text{if } n_2 < n_{in}^H \\ 1, & \text{if } n_2 = n_{in}^H \\ p \in [0, \bar{p}_4], & \text{if } n_2 > n_{in}^H \end{cases}$$

is the unique locally undefeated naturally separating equilibrium if and only if

$$d > (1 - \tau)(\eta_2 - p_0) \equiv d_2, \quad (21)$$

where $\bar{p}_4$ is defined in Online Appendix A.3.

Lemma 3 states that the two types will separate naturally when Condition (21) holds, meaning the highest early-follower count that the L -type can match is smaller than the H -type's informed followers. In other words, the L -type cannot afford to mimic a nonpurchasing H -type.

When the base price of fake accounts p_0 is higher than η_2 , Condition (21) holds regardless of d : *natural separation occurs even if antifake effort is zero*. This extreme case seems unrealistic given the prevalence of fake accounts. To rule out such an extreme case, we assume:

Assumption 3. $p_0 < \eta_2$.

4.2.3. Globally Undefeated Equilibrium. We note that the pooling, costly separating, and naturally separating equilibria may coexist. The following lemma summarizes the defeating relationship among these equilibria.

Lemma 4. *Given $d \leq d_1$ (so the pooling equilibrium exists), (a) the pooling equilibrium coexists with and defeats the costly separating equilibrium, and (b) the pooling equilibrium cannot coexist with the naturally separating equilibrium.*

Lemma 4 suggests that the pooling equilibrium may coexist with the costly separating equilibrium but always defeats it. The next proposition describes the globally undefeated equilibrium.

Proposition 1.

a. (No-pool) *If $d_1 \leq 0$, the unique globally undefeated equilibrium is*

$$\begin{cases} (n_2^{sep}, n_{in}^L), & \text{if } d \leq d_2 \quad (\text{costly separating}) \\ (n_{in}^H, n_{in}^L), & \text{otherwise} \quad (\text{naturally separating}) \end{cases}$$

b. (All-eqm) *If $d_1 > 0$, the unique globally undefeated equilibrium is*

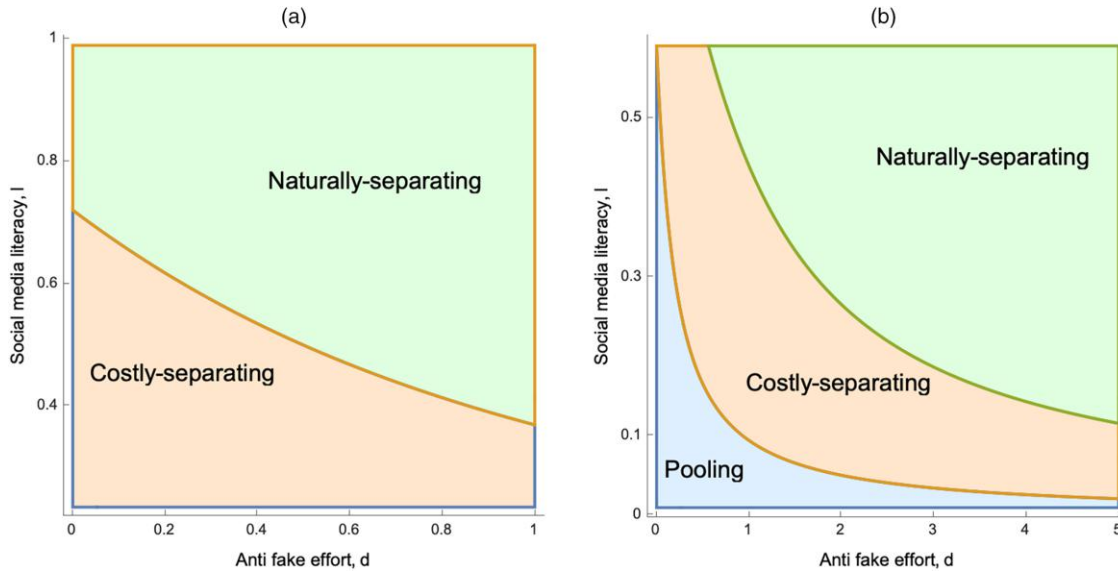
$$\begin{cases} (n_{in}^H, n_{in}^H), & \text{if } d \leq d_1 \quad (\text{pooling}) \\ (n_2^{sep}, n_{in}^L), & \text{if } d_1 < d \leq d_2 \quad (\text{costly separating}) \\ (n_{in}^H, n_{in}^L), & \text{if } d > d_2 \quad (\text{naturally separating}). \end{cases}$$

Proposition 1 suggests that the pooling equilibrium cannot exist when the fake account base price p_0 is relatively high (such that $d_1 \leq 0$) (case *No-pool*). In such a case, when the antifake effort d is relatively low, a costly separating equilibrium prevails; otherwise, a naturally separating equilibrium would prevail. Figure 3(a) illustrates such a case: with a relatively high fake account base price p_0 , as the antifake effort increases, the equilibrium regime transitions from costly separating to naturally separating. When the fake account base price p_0 is relatively low such that $d_1 > 0$, Figure 3(b) illustrates such a case (case *All-eqm*): the equilibrium transitions from pooling to costly separating and then to naturally separating as the antifake effort increases.

Proposition 1 and the preceding lemmas suggest that the influencer's equilibrium fake account purchasing behavior is discontinuous. Specifically, as the equilibrium transitions from pooling to costly separating, the L -type's offensive purchasing first increases and then suddenly drops to zero. The H -type's defensive purchasing first stays at zero and then suddenly jumps to a very high level. Such "rugged" equilibrium behaviors are further illustrated in the next subsection (see Figure 4).

4.3. Comparative Statics

Next, we conduct a set of comparative statistics on how the equilibrium fake account purchasing changes with the underlying parameters under different equilibrium regimes.

Figure 3. (Color online) Illustration of the Equilibrium Regime Transitions

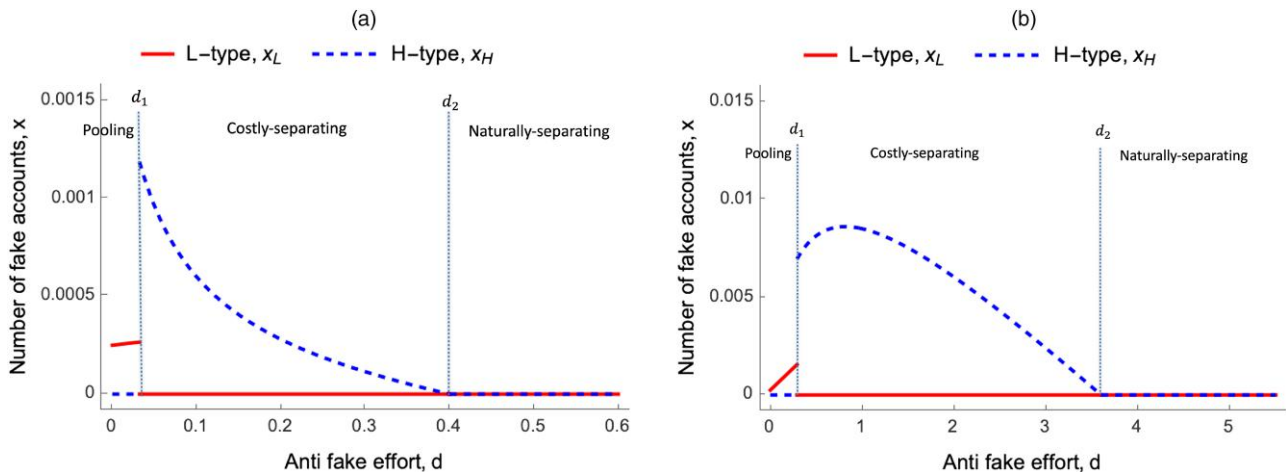
Notes. (a) No-pool ($p_0 = 2$). (b) All-eqm ($p_0 = 0.5$). $\mu = 10$, $\lambda_i = 0.3$, $\lambda_p = 0.3$, $\rho = 0.1$, $q_H = 20$, $q_L = 10$, $c_0 = 1$, $c_1 = 0.02$, and $\tau = 0.8$.

Proposition 2. Under the pooling equilibrium, the L -type's offensive purchase x_L^{pool} :

- increases in the platform's antifake effort d and the social media literacy l ,
- decreases in the antifake technology level τ and quality ratio R , and
- is unaffected by the fake account base price p_0 or the influencer's bargaining power λ_i .

The intuition for Proposition 2 is as follows: as the antifake effort increases or the technology level decreases, the consumer nuisance cost increases, causing more informed consumers to drop out. The L -type loses informed consumers more quickly than the H -

type,⁹ implying the gap between the two types' informed followers increases. Consequently, the L -type must buy more fake accounts to stay in the pooling equilibrium. This is illustrated in Figure 4 (the pooling regions). The intuition for the effect of social media literacy l is similar: increasing social media literacy leads to more informed followers for both types of influencers and also enlarges the gap between the two types' informed followers, which forces the L -type to buy more to stay in the equilibrium. Increasing the antifake technology level decreases the consumers' nuisance cost and shrinks the gap between the two types' informed followers. Increasing the quality ratio R reduces the

Figure 4. (Color online) Impact of Antifake Effort on the Number of Fake Accounts

Notes. (a) High antifake technology ($\tau = 0.9$). (b) Low antifake technology ($\tau = 0.1$). $\mu = 10$, $\rho = 0.12$, $l = 0.5$, $\lambda_i = 0.3$, $\lambda_p = 0.3$, $q_H = 20$, $q_L = 10$, $c_0 = 0.01$, $c_1 = 0.2$, $p_0 = 0.5$.

quality gap between the two types, resulting in the gap between the two types' informed followers. Consequently, the L -type needs fewer fake accounts to stay in the equilibrium. Finally, the number of fake accounts and the influencer's share of advertising revenue doesn't affect the gap between the two types' informed followers. Therefore, the L -type's purchase is unaffected by the fake account base price or the influencer's bargaining power.

Proposition 3. Under the costly separating equilibrium, the H -type's defensive purchase x_H^{csep}

- decreases in social media literacy l , fake account base price p_0 , quality ratio R , and antifake technology level τ ,
- increases in the influencer's bargaining power λ_i , and
- decreases in d if the following condition holds; otherwise, it first increases and then decreases in d .

$$c_1(1 - \tau)^2 \leq \frac{\lambda_i \mu(1 - l^2)c_0}{p_0[\lambda_i \mu(1 - l^2) - p_0 l]}. \quad (22)$$

The intuition for Proposition 3 is as follows. As seen from Equation (20), H -type's defensive purchasing is the highest number of informed followers the L -type is willing to mimic. Furthermore, x_H^{csep} increases in the L -type's net gain from mimicry $[\lambda_i \mu(1 - l^2) - p_f l]((c/q_L) - (c/q_H))$ and decreases in the price of fake accounts p_f . The former is determined further by the gap between the L - and H -type's followers $((c/q_L) - (c/q_H))$ and the L -type's marginal gain per follower from mimicry $[\lambda_i \mu(1 - l^2) - p_f l]$. Increasing social media literacy l decreases the marginal gain from mimicry (as there will be fewer uninformed consumers) and causes defensive purchasing to decrease. Increasing the fake account base price p_0 decreases L -type's marginal gain from mimicry and increases the fake account price. Both effects lead to decreased defensive purchasing. Increasing the quality ratio q_L/q_H reduces the gap between the two types' followers and L -type's marginal gain from mimicry and thus decreases the H -type's purchasing. Increasing the antifake technology level τ decreases the consumers' nuisance cost c and thus the gap between the two type's followers; it also increases the price of fake accounts p_f . Both reduced follower gap and increased price lead to decreased defensive purchasing. Increasing the influencer's bargaining power λ_i increases the L -type's marginal gain per follower and, thus, the H -type's defensive purchasing.

Increasing the antifake effort d can produce countervailing effects. As d increases, both the unit price of fake accounts p_f and the consumers' nuisance cost c increase. The former (the *higher fake account price effect*) reduces defensive purchasing, with a negative marginal effect of

$$m_1 = -\frac{1}{1 - \tau} \frac{\lambda_i \mu(1 - l^2) q_H - q_L}{p_f^2} c.$$

The latter (the *higher nuisance cost effect*) increases the L -type's marginal gain from mimicry and thus increases defensive purchasing, with a positive marginal effect of

$$m_2 = c_1(1 - \tau) \frac{\lambda_i \mu(1 - l^2) - p_f l q_H - q_L}{p_f q_H q_L}.$$

As d increases, fake account price p_f increases, and the higher nuisance cost effect $m_2 \rightarrow 0$. Therefore, as d increases, the negative effect m_1 will dominate eventually, causing defensive purchasing to decrease. When the technology level τ is relatively high such that (22) holds, the negative effect m_1 always dominates the positive effect m_2 , even for a low antifake effort. In such a case, the defense purchasing monotonically decreases, as illustrated in Figure 4(a) (the costly separating region). Otherwise, the positive effect dominates for low d but not for high d , causing the defensive purchasing to first increase and then decrease, as illustrated in Figure 4(b) (the costly separating region).¹⁰

Figure 5 illustrates the impact of social media literacy and technology level. In Figure 5(a), the L -type's offensive purchasing (in the pooling region) increases in social media literacy, as predicted by Proposition 2, whereas the H -type's defensive purchasing (costly separating) decreases, as predicted by Proposition 3. Figure 5(b) shows that increasing the antifake technology level can reduce both offensive and defensive fake account purchasing in their respective equilibrium regions.

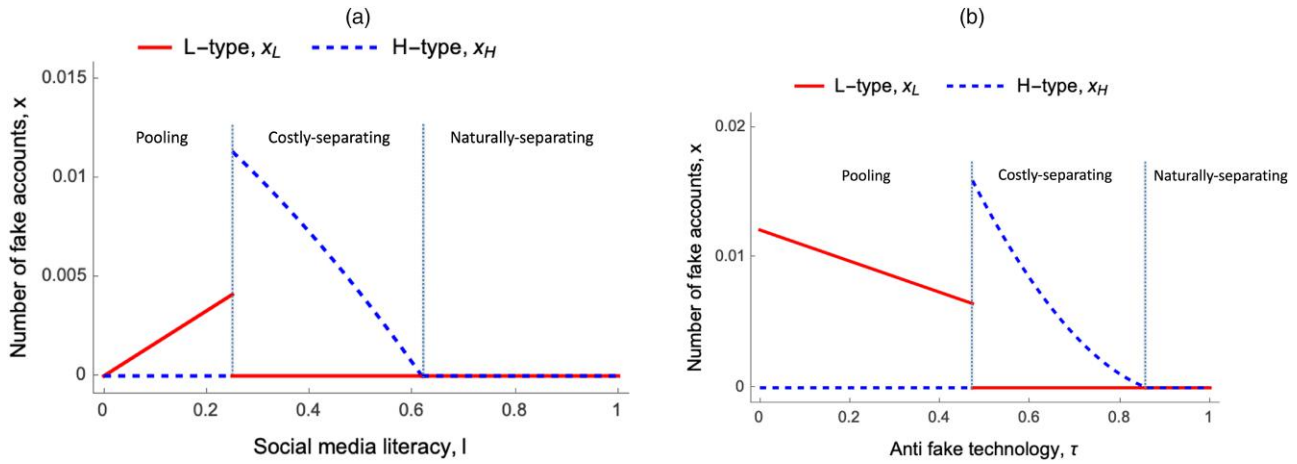
5. Antifake Effort and Consumer Welfare

We now turn our attention to the relationship between the platform's antifake effort and consumer welfare. We let $U^{pool}(d)$, $U^{csep}(d)$, and $U^{nsep}(d)$ denote consumer welfare under each equilibrium type, provided that the effort d sustains the equilibrium. The following lemma establishes the effect of antifake efforts on consumer welfare:

Lemma 5. $U^{pool}(d)$, $U^{csep}(d)$, and $U^{nsep}(d)$ monotonically decrease with antifake effort d .

Intuitively, under each equilibrium type, increasing the antifake effort would not impact consumers' inference of influencer quality but increase their nuisance costs, which is welfare reducing. Lemma 5 implies that consumers always prefer a costly separating equilibrium to a naturally separating one because the two equilibria provide the same information about influencer quality, but the latter imposes a higher nuisance cost.

Lemma 5 does not imply, however, that consumers would always prefer zero antifake effort. To see this, we note that, holding the nuisance cost constant, informed consumers are indifferent between equilibrium types, but uninformed ones prefer a separating equilibrium because it leads to better following decisions. When the benefit of separation outweighs the increase in nuisance

Figure 5. (Color online) Impact of Parameters on the Equilibrium Number of Fake Accounts

Notes. (a) Impact of social media literacy ($\tau = 0.5$, $d = 1$). (b) Impact of antifake technology level ($l = 0.1$, $d = 4$). $\mu = 10$, $\rho = 0.1$, $\lambda_i = 0.3$, $\lambda_p = 0.3$, $q_H = 20$, $q_L = 5$, $c_0 = 0.01$, $c_1 = 0.2$, and $p_0 = 1$.

costs, consumers, as a whole, may prefer a separating equilibrium.

Lemma 6. Consumers' preference for equilibrium types and antifake effort d^C is summarized in Table 2.

As seen in Table 2, when the pooling equilibrium is unavailable (*No-pool*), consumers prefer the costly separating equilibrium with zero antifake effort. When all three equilibria are available (*All-eqm*), they may prefer the pooling with zero antifake effort or the costly separating equilibrium with minimum antifake effort d_1 . The conditions for different cases are provided in the proof.

6. Platform's Optimal Strategy

Having examined consumers' preferred antifake effort, we now turn to the platform itself.

6.1. Platform's Optimal Antifake Effort

We denote $\pi_p^{pool}(d)$, $\pi_p^{csep}(d)$, and $\pi_p^{nsep}(d)$ as the platform's profit under the pooling, costly separating, and naturally separating equilibrium, respectively, provided that the chosen d sustains the equilibrium. Lemma 7 summarizes how the platform profit changes with the antifake effort.

Lemma 7. $\pi_p^{pool}(d)$, $\pi_p^{csep}(d)$, and $\pi_p^{nsep}(d)$ decrease in d .

The intuition behind Lemma 7 is as follows. In general, within each equilibrium type, increasing the antifake

effort affects the platform's profit in two ways. First, it increases consumer nuisance costs, reducing the number of participating consumers. This *consumer inconvenience* effect negatively affects the total advertising revenue and, thus, the platform's revenue share. Second, the platform incurs a higher cost for its antifake effort. Both effects reduce the platform's profitability.

Lemma 8.

- (*No-pool*) If $d_1 \leq 0$, $\pi_p^{csep}(0) > \pi_p^{nsep}(d_2)$
- (*All-eqm*) If $d_1 > 0$, (1) $\pi_p^{pool}(0) > \pi_p^{csep}(d_1)$, and (2) $\pi_p^{csep}(d_1) > \pi_p^{nsep}(d_2)$.

To see Lemma 8(b.1), we first note that the number of uninformed followers is a concave function of the expected quality of the influencer. Consequently, the number of uninformed followers under the pooling equilibrium, for an influencer with mean quality, is higher than the expected number of uninformed followers under a separating equilibrium, which is the average number of followers between an *H*- and an *L*-type influencer. Therefore, the advertising revenue (and the platform's share) is higher under a pooling equilibrium. This, together with the observation that the platform incurs an antifake cost under the separating equilibrium, suggests that the platform's profit is higher under the pooling equilibrium than under the separating one.

To see Lemma 8(a) and (b.2), we note that the two separating equilibria provide the same information about

Table 2. Consumer-Optimal Antifake Effort and Equilibrium Type

Equilibrium type	Consumers prefer	Consumer-optimal effort d^C	Condition
No-pool	Costly separating	0	—
All-eqm	Pooling	0	$U^{pool}(0) \geq U^{csep}(d_1)$
	Costly separating	d_1	$U^{pool}(0) < U^{csep}(d_1)$

Table 3. Platform-Optimal Antifake Effort and Equilibrium Type

Equilibrium type	Platform prefers	Platform-optimal effort d^*	Condition
No-pool	Costly separating	0	—
All-eqm	Pooling	0	$\Pi_p^{pool}(0) \geq \Pi_p^{csep}(d_1)$
	Costly separating	d_1	$\Pi_p^{pool}(0) < \Pi_p^{csep}(d_1)$

influencer quality, but the naturally separating equilibrium mandates a higher antifake effort, which results in fewer participating consumers and a lower revenue.

Proposition 4. *The platform’s optimal antifake effort and the associated equilibrium type are summarized in Table 3. Specifically, (a) a purely profit-driven platform (i.e., $w = 0$) optimally chooses zero antifake effort, whereas (b) a consumer-oriented platform ($w > 0$) may optimally choose a positive antifake effort to induce the costly separating equilibrium.*

Proposition 4(a) suggests that a purely profit-driven platform always chooses zero antifake effort. This is because, according to Lemma 8, its profit is maximized under the pooling equilibrium, which requires no antifake effort. Proposition 4(b) suggests a more consumer-oriented platform may prefer a separating equilibrium. This is because such an equilibrium may yield higher consumer welfare, which the platform values.

6.2. Comparative Statics

In this subsection, we examine the impact of three model parameters: social media literacy, fake account base price, and antifake technology level. We focus on these parameters because they hold strong managerial implications.

6.2.1. Comparative Statics on Consumer Welfare

Proposition 5. *The impact of parameters on consumer welfare is given in Table 4. Specifically, increasing social media literacy, fake account base price, and antifake technology level weakly improve consumer welfare.*

Consumer welfare loss comes from two sources: the nuisance cost of antifake efforts and suboptimal following decisions by uninformed consumers under the pooling equilibrium (or *cost of pooling*). Under the pooling equilibrium (Table 4, case 1), increasing social media literacy will reduce the cost of pooling because there are fewer uninformed consumers. Increasing fake account

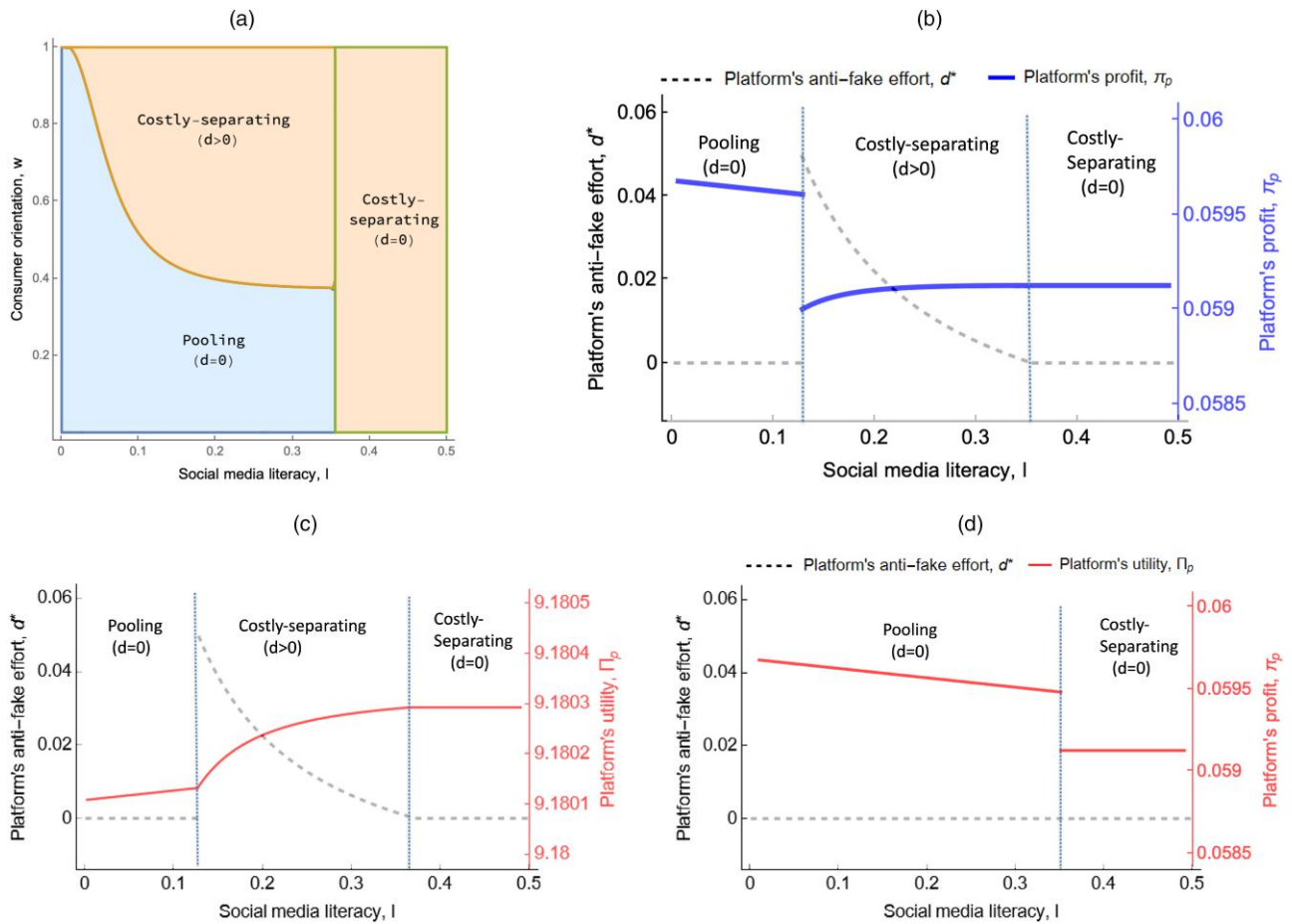
base price and antifake technology does not affect the cost of pooling or consumer welfare. Under the costly separating equilibrium with zero antifake effort (case 2), as the influencers are already separated and there is no nuisance cost from the antifake effort, neither parameter has any impact on consumer welfare. Finally, under the costly separating equilibrium with positive antifake effort (case 3), there is no cost of pooling, but the nuisance cost from the antifake effort decreases in social media literacy, fake account base price, and antifake technology; thus, consumer welfare increases in these parameters.

Next, we examine how each parameter affects the platform’s antifake effort, profit, and utility. We rely on numeric methods for these analyses because the boundary conditions outlined by Proposition 4 are not analytically tractable. We note that whereas the patterns we report in the next subsections are illustrated using a specific set of model parameters, they are fairly representative when we conduct robustness checks by systematically varying model parameters (available upon request).

6.2.2. Social Media Literacy. Figure 6 illustrates the effect of social media literacy (l) on the platform’s optimal effort, profit, and utility. Figure 6(a) shows that when l is moderate, a more consumer-oriented platform may optimally induce a costly separating equilibrium with positive antifake effort, whereas a more profit-focused platform may prefer a pooling equilibrium with zero antifake effort. The higher the l , the lower the minimum consumer orientation required for the separating equilibrium, indicating that the platform is more likely to exert antifake effort. When l is low enough, consumer welfare is higher under the pooling equilibrium so that even the most consumer-oriented platform would prefer the pooling equilibrium. When l reaches a high level (≥ 0.355 in this example), the pooling equilibrium no longer exists, and the platform optimally induces a

Table 4. Comparative Statistics for Consumer Welfare with Platform-Optimal Antifake Effort

Parameters	Impact on consumer welfare		
	(1) Pooling with $d^* = 0$	(2) Costly separating with $d^* = 0$	(3) Costly separating with $d^* > 0$
Social media literacy	↑	=	↑
Fake account base price	=	=	↑
Antifake technology level	=	=	↑

Figure 6. (Color online) Impact of Social Media Literacy on the Platform

Notes. (a) Platform's optimal strategy. (b) Platform's optimal effort and profit ($w = 0.5$). (c) Platform's optimal effort and utility ($w = 0.5$). (d) Platform's optimal effort and utility ($w = 0.1$). $\mu = 0.2$, $\rho = 0.3$, $\lambda_i = 0.3$, $\lambda_p = 0.3$, $p_0 = 0.1$, $q_H = 100$, $q_L = 10$, $c_0 = 0.2$, $c_1 = 0.02$, $\tau = 0.8$, and $\gamma = 0.1$.

costly separating equilibrium with zero antifake effort, regardless of its consumer orientation.

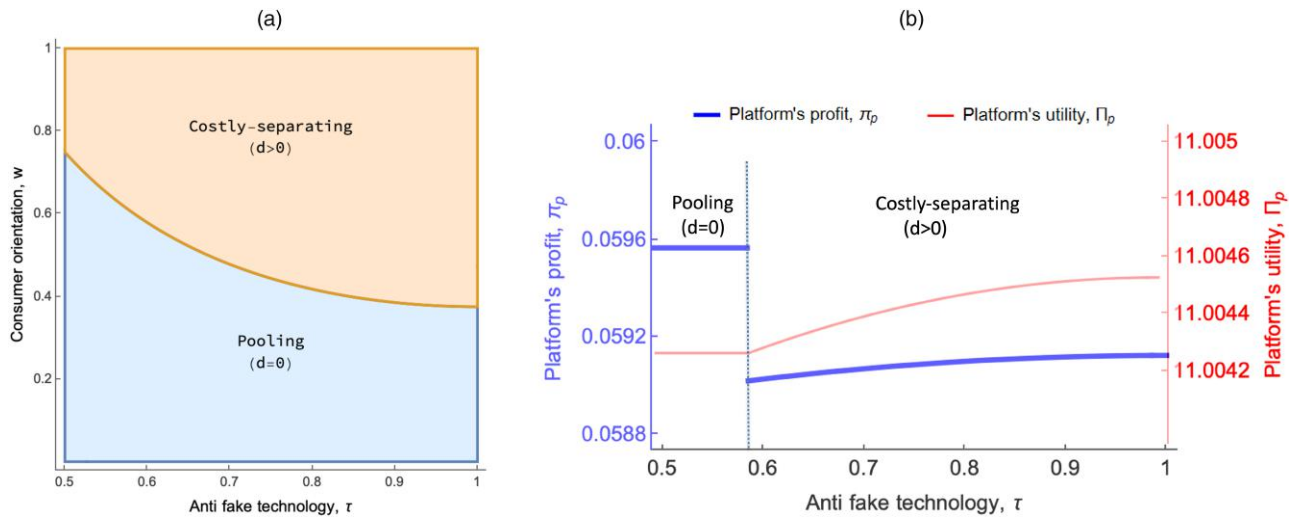
Figure 6(b) further confirms that the platform's anti-fake effort may not be monotonic in social media literacy; a consumer-oriented platform may prefer zero antifake effort when social media literacy is low or very high. Furthermore, when the platform does exert a positive antifake effort, its effort level decreases as social media literacy increases, suggesting social media literacy education can substitute for the platform's antifake effort. The figure also shows that the platform's profit is the highest with zero social media literacy.¹¹

Figure 6, (c) and (d) shows that the platform's preferred social media literacy depends on its consumer orientation. Recalling that consumer welfare increases with social media literacy, when the platform is highly consumer oriented, it has sufficient concern for consumer welfare that it may benefit from a higher social media literacy, as shown in Figure 6(c). Conversely, a more profit-focused platform may prefer a lower social media literacy, as illustrated in Figure 6(d).

6.2.3. Fake Account Base Price. Our findings about the effect of fake account base price are similar to those about social media literacy (see Figure 8 of Online Appendix A.15). In particular, (a) when the fake account base price is relatively low, a more profit-driven platform would not invest in any antifake efforts, but a more consumer-oriented platform may; (b) when the fake account base price is high enough, pooling equilibrium no longer exists, and the platform optimally chooses a costly separating equilibrium with zero antifake effort, regardless of its consumer orientation; and (c) the platform may prefer a lower fake account base price when it is more profit focused; the reverse may be true when it is sufficiently consumer oriented.

6.2.4. Antifake Technology Level. Figure 7 illustrates the effect of the antifake technology level. Figure 7(a) shows that as the technology level increases, the minimum consumer orientation to sustain a separating equilibrium decreases, suggesting that better antifake technology makes antifake efforts more likely. Figure 7(b) also shows

Figure 7. (Color online) Impact of Antifake Technology on the Platform’s Optimal Antifake Effort and Profit



Notes. (a) Platform’s optimal strategy. (b) Platform’s profit and utility ($w = 0.6$). We let $\mu = 0.2$, $\rho = 0.3$, $\lambda_i = 0.3$, $\lambda_p = 0.3$, $l = 0.2$, $p_0 = 0.1$, $q_H = 100$, $q_L = 10$, $c_0 = 0.2$, $c_1 = 0.02$, $\tau = 0.8$, and $\gamma = 0.1$.

that the platform is unresponsive to changes in antifake technology under a pooling equilibrium, regardless of its consumer orientation. When the technology level is high enough, a consumer-oriented platform may prefer a costly separating equilibrium. In such a case, further improvements in the antifake technology would increase the platform’s utility and profitability and reduce its antifake effort (not plotted). Overall, the platform weakly benefits from better antifake technology.

6.3. Comparing Platform and Consumer Optima

Proposition 6. *The relationship between the platform and consumer optima is given in Table 5. In general, the platform exerts weakly less antifake effort than what is optimal for consumers.*

As seen in Table 5, in some cases (1, 2, and 4), the platform’s optimal antifake effort is the same as the consumer’s. Among these, case 4 exists only when the platform is sufficiently consumer oriented. In case 3, the platform’s optimal antifake effort is less than the consumers’. Such a case is more likely when the platform is more profit driven. In such a case, consumers prefer a separating equilibrium with positive antifake effort, whereas the platform prefers a pooling equilibrium with zero antifake effort.

7. Extensions

7.1. Three Types of Influencers

Our main model assumes only two types of influencers. One may wonder whether our insights are generalizable to more influencer types. To address this issue, we extend our model to three influencer types. Specifically, we assume an influencer’s quality is drawn randomly from three levels, $q_L < q_M < q_H$, with probabilities ρ_L, ρ_M , and ρ_H ($\rho_H = 1 - \rho_M - \rho_L$), respectively. The following proposition summarizes the equilibrium types and fake account purchasing patterns in this extended model.

Proposition 7. *With three influencer types, the unique undefeated equilibrium is given in Table 6 (the corresponding beliefs and conditions are provided in Online Appendix B).*

As seen in Table 6, the three types of equilibria—pooling, costly separating, and naturally separating—are preserved and become the *fully pooling*, *costly fully separating*, and *naturally fully separating* equilibria, respectively. We also obtain two new “hybrid” equilibrium types where two types pool and separate from the third type: *hybrid with H-M pooling* and *hybrid with M-L pooling*. Similar to the main model, an influencer may purchase fake accounts defensively (e.g., *H-* and *M-*

Table 5. Consumer-Optimal Antifake Effort d^C vs. Platform-Optimal d^*

Scenario		Platform optima		Consumer optima		Conditions
		Eqm	d^*	Eqm	d^C	
No-pool	1	csep	0	csep	0	—
All-eqm	2	pool	0	pool	0	$U^{csep}(d_1) \leq U^{pool}(0)$
	3	pool	0	csep	d_1	$U^{csep}(d_1) > U^{pool}(0)$ & $\Pi_p^{pool}(0) \geq \Pi_p^{csep}(d_1)$
	4	csep	d_1	csep	d_1	$U^{csep}(d_1) > U^{pool}(0)$ & $\Pi_p^{pool}(0) < \Pi_p^{csep}(d_1)$

Table 6. Undeclared Equilibrium with Three Types of Influencers

Equilibrium type	Notation	H-type	M-type	L-type
Costly fully separating	$H^* M^* L$	$\frac{\lambda_i \mu (1-l^2)(n_r^H - n_r^L)}{p_f} + n_{in}^L$	$\frac{\lambda_i \mu (1-l^2)(n_r^M - n_r^L)}{p_f} + n_{in}^L$	n_{in}^L
Naturally fully separating	$H M L$	n_{in}^H	n_{in}^M	n_{in}^L
Fully pooling	HM^*L^*	n_{in}^H	n_{in}^H	n_{in}^H
Hybrid with M-L pooling	$H ML^*$	n_{in}^H	n_{in}^M	n_{in}^M
Hybrid with H-M pooling	$HM^* L$	n_{in}^H	n_{in}^H	n_{in}^L

Note. * denotes fake account purchasing, and “|” denotes separation between the two adjacent influencer types.

types purchase fake accounts under the costly fully separating equilibrium) or offensively (e.g., M - and L -types under the fully pooling equilibrium) in this extended model. Different from the main model, the M -type's purchase may be simultaneously defensive and offensive (e.g., hybrid with H-M pooling). In addition, under two equilibrium scenarios (i.e., costly fully separating and fully pooling), two of three influencer types purchase fake accounts. This suggests that the extended model can capture the case where fake account purchasing is prevalent and occurs at multiple levels of influencer quality.

7.2. A Repeated Game

Another potential concern is whether the insights from a one-shot game could be generalized to a repeated setting. When the game is repeated, some existing consumers may remain, and new consumers may join. One may expect that with repetition, consumers, on average, could become more informed: they may learn the influencer's quality from prior experience with the influencer or from prior revelation of the influencer type (by a separate equilibrium); they may also learn from prior consumers through word of mouth. On the other hand, information asymmetry may linger because existing uninformed consumers may remain, and new uninformed consumers may join. We similarly expect that the information asymmetry encountered by advertisers will gradually diminish but not disappear immediately. In the following, we use a simplified two-period game to examine how repetition may impact equilibrium behavior and whether the insights from the one-shot game can be applied in the repeated setting.

We consider a repeated game where our original game is repeated for another period, with the influencer type persistent across the two periods. For simplicity, we assume that consumers, fake accounts, and the advertiser are short-lived (i.e., they live for only one period), and the number of consumers stays constant across two periods. To capture the notion that information asymmetry may decrease but will not disappear completely, we assume that the proportion of informed

consumers in period 2, l_2 , is higher than in period 1 (l) and is a function of the equilibrium type in period 1:

$$l_2 = \begin{cases} l + \delta_{sep} \equiv l_2^{sep}, & \text{if period-1} \\ & \text{equilibrium} \\ & \text{is separating} \\ l + \delta_{pool} \equiv l_2^{pool}, & \text{if period-1} \\ & \text{equilibrium} \\ & \text{is pooling} \end{cases}, \text{ where } \delta_{sep} > \delta_{pool} > 0.$$

We interpret consumers in period 2 as a combination of remaining consumers from period 1 and newly joined consumers in period 2. We interpret fake accounts in period 2 as a sum of renewed existing fake accounts and newly purchased fake accounts (if any).

For simplicity, we assume the probability of drawing an informed advertiser in each period is the same as the proportion of informed consumers (relaxing this does not fundamentally change our result). Thus, the advertiser also becomes more informed but may not be fully informed in period 2.

We assume no discounting and that the platform's antifake effort remains the same across the two periods; that is, the platform chooses its effort once at the beginning of period 1. For simplicity, we also assume that the influencer only considers the short-term impact of their current-period deviation: they do not consider any impact of their current-period deviation on the future period. The following proposition describes the equilibrium types across the two periods.

Proposition 8.

a. (No-pool in period 1) If $d_1 \leq 0$, the equilibrium types in two periods are

$$\begin{cases} (\text{costly separating}, & \text{if } d \leq d'_2(l_2^{sep}) \\ \text{costly separating}), & \\ (\text{costly separating}, & \text{if } d'_2(l_2^{sep}) < d \leq d_2 \\ \text{naturally separating}), & \\ (\text{naturally separating}, & \text{if } d > d_2. \\ \text{naturally separating}), & \end{cases}$$

b. (All-*eqm* in period 1) If $0 < d_1 \leq d'_2(l_2^{pool})$, the equilibrium types in two periods are

$$\left\{ \begin{array}{ll} (\text{pooling}, \text{pooling}), & \text{if } d \leq d'_1(l_2^{pool}) \\ (\text{pooling}, \text{costly separating}), & \text{if } d'_1(l_2^{pool}) < d \leq d_1 \\ (\text{costly separating}, & \text{if } d_1 < d \leq d'_2(l_2^{sep}) \\ \text{costly separating}), & \\ (\text{costly separating}, & \text{if } d'_2(l_2^{sep}) < d \leq d_2 \\ \text{naturally separating}), & \\ (\text{naturally separating}, & \text{if } d > d_2. \\ \text{naturally separating}), & \end{array} \right.$$

c. (All-*eqm* in period-1) If $d_1 > d'_2(l_2^{pool})$, the equilibrium types in two periods are

$$\left\{ \begin{array}{ll} (\text{pooling}, \text{pooling}), & \text{if } d \leq d'_1(l_2^{pool}) \\ (\text{pooling}, \text{costly separating}), & \text{if } d'_1(l_2^{pool}) < d \leq d'_2(l_2^{pool}) \\ (\text{pooling}, \text{naturally separating}), & \text{if } d'_2(l_2^{pool}) < d \leq d_1 \\ (\text{costly separating}, & \text{if } d_1 < d \leq d'_2(l_2^{sep}) \\ \text{costly separating}), & \\ (\text{costly separating}, & \text{if } d'_2(l_2^{sep}) < d \leq d_2 \\ \text{naturally separating}), & \\ (\text{naturally separating}, & \text{if } d > d_2. \\ \text{naturally separating}), & \end{array} \right.$$

where $d'_1(\cdot)$ and $d'_2(\cdot)$ are defined similarly as d_1 and d_2 (see Online Appendix C for details).

Proposition 8 shows that, given the platform's antifake effort in period 1, the equilibrium may stay the same in period 2 or transition to a more advanced equilibrium (i.e., pooling $\rightarrow$ costly separating, pooling $\rightarrow$ naturally separating, or costly separating $\rightarrow$ naturally separating). Therefore, the separating (pooling) equilibrium could become more (less) prevalent in the long run, which is a result of reduced information asymmetry.

An influencer may purchase fake accounts in both periods (e.g., the *L*-type under (pooling, pooling)), just one (e.g., the *L*-type under (pooling, costly separating)), or none (e.g., the *L*-type under (costly separating, costly separating)). In the case of (pooling, pooling), the *L*-type would purchase more fake accounts in period 2 according to Proposition 2, noting that $l_2 > l$. In the case of (costly separating, costly separating), the *H*-type would purchase fewer in period 2 by Proposition 3. In general, our one-shot game corresponds to a stage game in the repeated setting. Consequently, our earlier insights into different equilibrium types can be leveraged to understand how the system will evolve in the repeated setting.

8. Discussion and Conclusion

Motivated by the prevalence of social media fake accounts in the influencer economy and a lack of understanding of

this phenomenon, we study a fake account model in which influencers can purchase fake accounts to make them appear more popular to consumers and advertisers, whereas the social platform can mount an antifake effort that increases the cost of fake accounts while also increasing the nuisance cost of consumers. We use this model to study the influencer's equilibrium fake account purchasing behavior, the platform's optimal antifake effort, consumer welfare, and how the ecosystem responds to changes in several model parameters.

8.1. Contribution to the Literature

Our paper contributes to the literature in three main ways. First, we contribute to the understanding of fake account purchasing behaviors in the influencer economy. We find that not only low-quality influencers may purchase fake accounts to mimic high-quality influencers, but high-quality influencers may also purchase fake accounts to separate themselves. The latter type of equilibrium has received little attention in the literature on deceptive behaviors but is practically relevant and holds important implications for how we view and tackle the problem of fake accounts. We also show that as the platform's antifake effort increases, the equilibrium regime generally transitions from a pooling equilibrium, where the low-quality influencer purchases fake accounts to mimic a high-type influencer, to a costly separating one, where the high-quality influencer purchases fake accounts to prevent a pooling equilibrium, and to a naturally separating one, where the two types separate without purchasing fake accounts. Interestingly, the system behaves erratically in the sense that the number of fake accounts may increase and even rise sharply in response to increased antifake efforts. For example, the number of fake accounts can increase with the antifake effort before eventually decreasing; it may also jump significantly as the equilibrium regime transitions from pooling to separating.

Second, we enhance our understanding of how social media platforms may choose their level of antifake efforts and the potential impact of these choices on consumer welfare. We find that a purely profit-driven platform lacks incentive to implement antifake efforts. Even if the platform becomes more consumer oriented, it generally exerts less antifake effort than what is optimal for consumers. Consumers may prefer a costly separating equilibrium with a positive antifake effort, whereas a profit-driven platform tends to prefer a pooling equilibrium with zero antifake effort. The misalignment arises because the platform can attract more uninformed followers (resulting in higher advertising revenue) under a pooling equilibrium than under a separating equilibrium, whereas consumers may prefer the latter.

Finally, we offer novel insights into the effects of different antifake measures. These include platform-led antifake initiatives, improving consumers' social media

literacy, increasing the cost of fake accounts (e.g., through stronger fake account laws), and advancing more effective antifake technologies. We show that whereas positive antifake efforts may be necessary for sustaining a separating equilibrium, neither the platform nor consumers can benefit from additional antifake efforts beyond what is necessary for sustaining a desired equilibrium. This is because additional antifake efforts increase consumer nuisance costs and may not always reduce information asymmetry about influencer quality.

Consistent with our findings on the effect of antifake effort, the system exhibits erratic behavior in response to other antifake measures: for example, under the pooling equilibrium, the number of fake accounts may increase in social media literacy and not respond to changes in the fake account base price. Moreover, a profit-focused platform may prefer a lower social media literacy and a lower fake account base price, even though consumers can benefit from higher social media literacy and a higher fake account base price. In contrast, both consumers and the platform are weakly better off with better antifake technology.

8.2. Managerial Implications

Our findings hold managerial implications for platforms, policymakers, and consumer protection agencies. For platforms, one important insight from our analyses is that platforms should not measure the success of their antifake efforts using the resulting number of fake accounts. This is because the equilibrium number of fake accounts may increase and even jump as the platform exerts more antifake efforts and because fake accounts can also be beneficial: high-quality influencers may also purchase fake accounts to separate themselves. Instead, platforms should focus on whether their antifake effort allows consumers and advertisers to better differentiate between high- and low-quality influencers. The same also holds for policymakers and regulators who hope to measure the success of their policies and regulations.

Second, the platform should not impose more than necessary antifake efforts. Our results show that additional antifake efforts can be costly for both the platform and consumers because of added nuisance costs and operating costs of antifake efforts.

For policymakers and regulators, we show that platforms generally underinvest in antifake efforts at the expense of consumer welfare. This is because antifake efforts can expose low-quality influencers and reduce platforms' user bases. Moreover, profit-focused platforms may lack incentives to promote social media literacy or implement measures that aim to increase fake account costs, although such measures can increase consumer welfare. In contrast, advancements in antifake technologies can benefit both platforms and consumers.

Consequently, policymakers and regulators should focus on incentivizing platform investments in advancing antifake technologies.

8.3. Limitations and Future Work

As an initial step in understanding the issue of fake accounts and developing coping strategies, we have specifically examined a type of fake accounts: those created to boost influencers' popularity. Our findings may not generalize to other types of fake accounts, such as those created for spreading scams, malware, and identity theft, or politically motivated fake accounts. Our model assumes that fake accounts distort consumers' and advertisers' perceptions of influencer quality but do not directly harm them. When fake accounts directly harm consumers, we anticipate that the platform will have stronger incentives to implement antifake measures. However, we also expect that many of the driving forces outlined in our model will continue to be relevant. Finally, our model follows the signaling game to assume one representative influencer that can be either low or high quality. Future research can also extend this model to directly incorporate competition between influencers.

Endnotes

¹ See <https://www.aspire.io/blog/influencer-budget-calculator>.

² For instance, Facebook/Instagram builds a "Creator Marketplace" to match influencers and brands with a project. It enables influencers to set up profiles, get discovered by brands with a good fit, and negotiate the content details for products/services. The influencer needs to clearly label the content of the product with a tag indicating the partnership with a brand and gets paid when sharing the content of products/services with followers.

³ Although we use the term "follow" here, the decision can also be interpreted as a subscription decision or a friendship request, provided that a positive decision allows the consumer to receive the influencer's content.

⁴ The driving force of the model remains the same if the informed consumers were only "imperfectly informed" (i.e., they do not know the true quality of the influencer but receive an informative but imperfect signal about it). The key assumption here is that some consumers are *better informed* than others.

⁵ We note that, although we use the number of followers as a quality signal, our model can be generalized to other quality signals such as the number of likes and the number of comments. This is because, just as an influencer can purchase fake followers, fake likes/comments generated by fake accounts can also be purchased.

⁶ We have also explored an alternative scheme where two or more advertisers compete for the ad slot via a sealed-bid second-price auction, with the auction payment split between the influencer and the platform. The results are quite similar because the driving forces of the model are still the same, although some analyses become less tractable.

⁷ To see this, note that if an H -type influencer has more (the same, fewer) early followers, they will also have more (the same, fewer) total followers.

⁸ In our context, an undefeated equilibrium requires that if an off-equilibrium action in the equilibrium is played by an t -type ($t \in \{H, L\}$) influencer in an alternative equilibrium and both types

are better off (at least one type is strictly better off) in the alternative equilibrium, then the belief for the off-equilibrium action in the focal equilibrium should be identical to the belief for the same action under the alternative equilibrium. If the equilibrium fails to hold under such an off-equilibrium belief, we say the equilibrium is *defeated* by the alternative equilibrium.

⁹ This is because informed followers have lower valuations for the *L*-type's content and thus are more likely to drop out when the influencer is the *L*-type (see Equation (8)).

¹⁰ As we normalize the number of consumers to a unit mass, the number of fake accounts in the figures (i.e., shown on the vertical axis) should be interpreted relatively. For instance, if the number of fake accounts is 0.03, we infer that there are 30 fake accounts per 1,000 consumers.

¹¹ This finding is robust with different model parameters, including different values of consumer orientation *w*.

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